Prodigy Anesthesia Quick Review

Scoliosis

What is scoliosis?	Scoliosis is a lateral curvature in the vertebra with a resulting deformity of the ribcage. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 904.*
What is kyphosis?	Kyphosis is a posterior curvature in the spinal column. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 653.*
What are the five categories of scoliosis and which is the most prevalent?	The five categories of scoliosis are based on their etiology. They are: idiopathic, congenital, neuropathic, myopathic, and traumatic. Neuropathic scoliosis is caused by diseases such as polio, cerebral palsy, and syringomyelia. Myopathic scoliosis is caused by conditions such as muscular dystrophy and amyotonia. Idiopathic scoliosis is the most prevalent form. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 653.*
What degree of spinal curvature is considered severe in patients with scoliosis?	A spinal curvature greater than 40 degrees is considered severe scoliosis. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 452-453.*
What preoperative diagnostic tests should be performed in patients with severe scoliosis?	Pulmonary function tests, arterial blood gases, and ECG should be performed in patients with severe scoliosis. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 904.*
What cardiovascular abnormalities typically accompany severe scoliosis?	Mitral valve prolapse is the most common cardiac abnormality associated with scoliosis. Pulmonary hypertension leading to right ventricular hypertrophy is a common sequela as progressive hypoxemia results in pulmonary vasoconstriction. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 654.*
What respiratory complications can accompany kyphosis and scoliosis?	The severe chest wall deformity can result in a restrictive respiratory pattern with a reduction in lung volumes with vital capacity, total lung capacity, functional residual capacity, and residual volume all reduced. V/Q mismatching with hypoxemia can develop. Cor pulmonale is a common finding. Although most patients are normally normocarbic, and elevated PaCO2 heralds impending ventilatory failure. The vital capacity is typically reduced to 60-80% of the predicted value. The FEV1/FVC is normal unless the patient's condition is complicated by concomitant pulmonary disease. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 653-654.*

Copyright 2015 Prodigy Anesthesia, Inc. Unlawful to duplicate, distribute, or sell this document.

Prodigy Anesthesia Quick Review

Scoliosis

Why might the use of nitrous oxide be discouraged during the anesthetic for patients with severe scoliosis?	Nitrous oxide may predispose the patient to increases in pulmonary vascular resistance and could worsen pulmonary hypertension. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 462-463.*
What treatment options are available for scoliosis?	Scoliosis may warrant observation without medical treatment, the use of braces or electronic stimulator, and anterior or posterior spinal fusion with rods. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 654.*

Copyright 2015 Prodigy Anesthesia, Inc. Unlawful to duplicate, distribute, or sell this document.